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AI/ML-Based Accessibility & Assistive Technology for Inclusive Healthcare

TRACK:

AI for Social Good

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Abstract:-

Healthcare accessibility is a concern of equity, human rights, and dignity; it is not just a technical issue. In the context of healthcare systems, millions of senior patients and people with disabilities still face challenges. The blind can't read prescriptions, the deaf can't communicate without interpreters, and senior patients often feel excluded from modern digital systems. In this paper, we describe a human-centric AI/ML system that leverages generative AI-driven voice assistant, OCR prescription reading with text-to-speech (TTS), and sign language recognition. This approach combines multiple assistive systems into one comprehensive healthcare system, as opposed to the fragmented systems that exist today. We describe the approach, discuss related research, and discuss what such a system might mean to patients and caregivers.

Keywords:- Accessibility, Assistive Technology, Inclusive Healthcare, Artificial Intelligence, Machine Learning, Sign Language Recognition, OCR, Voice Assistant

I. Introduction:-

Accessibility is really important for everyone. It is not about making things easy to use. It is about being fair and respectful to all people. Many people, with disabilities have a time doing things that other people do every day. For example a deaf person may not be able to join in a group conversation unless there is someone who can sign for them. A student who is blind may need someone to read to them. An older person or someone who has trouble moving around may have a time using computers or doing things on their own. These problems are not just because of technology. They are also because of the way society is set up. A lot of things were not designed to include everyone [1] [2].

Assistive technologies have been trying to fill these gaps, but many conventional technologies are not flexible enough. It has been seen that although technologies are available for the visually impaired, they are not accepted because of poor usability and lack of flexibility, especially when it comes to children [8]. Hearing-impaired people also have fragmented solutions for sign language interpretation, which are not scalable or available in real-time [5]–[7].

However, Artificial Intelligence (AI) and Machine Learning (ML) are now transforming this scenario. AI-based smart vision glasses are assisting visually impaired persons in reading and understanding their surroundings [3]. Sign language recognition models, especially those based on deep learning and Transformers, have shown incredible progress in converting signs into text and speech [6], [7]. Multi-agent systems are being developed to assist interactively, allowing both visually impaired and senior citizens to perform tasks independently [4]. Such developments indicate that AI, when designed carefully, can be an excellent facilitator of accessibility.

However, much of the current progress is still fragmented. Technologies are often addressing one specific problem but are not addressing the issue of how they can integrate with the overall ecosystem that can support individuals in various aspects of their lives. As Giansanti and Pirrera suggest [1], AI in accessibility needs to be driven by inclusivity and equity to ensure that it does not perpetuate

existing barriers.

This paper proposes a multi-modal, AI/ML-driven assistive framework that unifies sign language recognition, visual assistive technologies, and multi-agent conversational systems. Unlike siloed tools, this ecosystem aims to provide seamless, real-world accessibility that adapts to users' needs rather than forcing users to adapt to technology.

II. Literature survey:-

Sr. No.	Author(s) & Year	Research Focus	Methodology/Technique Used	Key Contributions	Limitations/Gaps Identified
1.	Giansanti & Pirrera (2025) [1]	AI and assistive technologies in healthcare	Narrative review of integrating AI in healthcare accessibility	Emphasizes equity and human-centered design.	Lacks specific technical recommendations; theoretical rather than implementation-focused
2.	Umucu (2025) [2]	Disability-inclusive AI design	Integrated framework for inclusive design	Proposes framework involving disabled individuals throughout design process	Theoretical framework requires implementation validation; lacks empirical testing
3.	Udayakumar et al. (2023) [3]	AI-powered smart vision glasses	OCR integration with contextual awareness; real-time auditory feedback	Enables prescription reading and navigation for visually impaired.	Long-term adoption depends on usability improvements; limited discussion of user testing
4.	Oliveira et al. (2022) [4]	Multi-agent assistance for visually impaired and elderly	Multi-agent interaction frameworks with contextual assistance	Supports reminders, safety monitoring, and independence through coordinated agent interactions	Personalization and cultural adaptation remain challenging; needs broader user testing
5.	Cherniavsky et al. (2008) [5]	Sign activity detection for mobile telecommunication	Activity detection algorithms for conversational sign language video	Enabled efficient video transmission by detecting signing	Limited to activity detection only, not full recognition or translation

				activity; foundational work for mobile SLR	
6.	Camgöz et al. (2020) [6]	End-to-end sign language recognition and translation	Sign Language Transformers with encoder-decoder architecture	Joint learning of recognition and translation.	Requires large-scale datasets; high computational complexity
7.	Huang et al. (2018) [7]	Video-based sign language recognition	Deep learning without temporal segmentation	Practical real-time recognition; reduces preprocessing requirements	Limited accuracy on complex, continuous signing; challenges with signer independence
8.	Gori et al. (2016) [8]	Review of devices for visually impaired people	Critical analysis of existing assistive technologies	Identifies gap between technological sophistication and user acceptance.	Only identifies problems without proposing solutions; dated but still relevant findings
9.	Bagwan et al. (2023)[9]	Sign language to speech translation	TensorFlow Object Detection API with transfer learning	Real-time gesture detection with text and audio output.	Limited to specific gesture set; needs more training data for expansion
10.	Chaturvedi & Ramaiya (2025)[10]	ISL real-time recognition using YOLOv11	YOLOv11 with MediaPipe hand pose estimation; 21 keypoint landmark annotation	Exceptional performance (97.75% precision, 95.002% recall, 97.635% mAP);	Focused primarily on alphabet gestures; needs expansion to words and continuous signing
11.	Kishor et al. (2026)[11]	Comprehensive review of ISL translation systems	Analysis of 20 significant ISL translation experiments; machine learning and deep learning focus	Traces evolution from rule-based to ML/DL approaches.	Review paper without original experimentation; identifies gaps but doesn't fill them
12.	Yadav et al. (2024)[12]	ISL image generation using GANs	Hybrid GAN combining ProGAN with Self-Attention; WGAN-GP loss function.	Generates high-quality, feature-rich ISL images.	Focuses on static image generation only.

Research on Artificial Intelligence and Machine Learning based assistive technology covers a lot of areas. These areas include sign language recognition, vision assistive devices and systems that help people in their lives. This section is a review of the research papers that have been done on these topics.

1. Sign Language Recognition and Translation Systems

Cherniavsky and his team did some work on sign language in 2008. They made a system that could detect when someone was using sign language in a video. This was a deal because it meant that sign language could be used on mobile phones.

Later Camgöz and his team made a system called Sign Language Transformers in 2020. This system can. Translate sign language at the same time. It is an improvement over older systems because it does not need someone to manually separate the different parts of the sign language.

Huang and his team made a system in 2018 that can recognize sign language from videos without needing to separate the parts. This makes the system more practical for life.

Bagwan and his team made a system in 2023 that can translate sign language into language. They used a tool called the TensorFlow Object Detection API to make the system.

Chaturvedi and Ramaiya made a system in 2025 that can recognize Sign Language. They used tools like MediaPipe and YOLOv11 to make the system.

Kishor and his team reviewed systems that can recognize Indian Sign Language in 2025. They found that these systems have improved a lot over the years.

Yadav and his team made a system in 2024 that can generate images of Sign Language. They used tools like Generative Adversarial Networks to make the system.

2. Vision Assistive Devices

Udayakumar and his team made glasses in 2023 that can help people who are blind. These glasses can read prescriptions. Help people navigate their surroundings.

Gori and his team reviewed devices that can help people who are blind in 2016. They found that many of these devices are not used by people because they are not easy to use.

3. Multi-Agent and Conversational Systems

Oliveira and his team made a system in 2022 that can help people who're blind and old. The system uses agents that work together to help people.

Giansanti and Pirrera wrote about how important it's to make assistive technology that is fair and helpful to everyone in 2025.

Umucu made a framework in 2025 that can help make technology that is fair and helpful to everyone.

There are gaps in the research that has been done so far. These gaps include:

1. There are not systems that can recognize sign language and help people who are blind at the same time.
2. There are not datasets for Indian Sign Language.
3. It is hard to recognize sign language in time.
4. It is hard to tell the difference between signs.

This review of the research that has been done far will help us make a new system that can recognize sign language help people who are blind and talk to people. Our system will be different from systems because it will be able to do all of these things at the same time. It will be able to adapt to the needs of the people who use it, than the other way around.The proposed system focuses on real-time sign language detection using the TensorFlow Object Detection API implemented in Python. Contrary to traditional recognition techniques, which are based on handcrafted features, our technique utilizes deep learning-based object detection models that are capable of recognizing hand gestures dynamically from video streams. The methodology is outlined below:

III. Proposed Methodology:-

The research and analysis performed above highlights the framework for comparative analysis to assess the efficiency of different Convolutional Neural Network (CNN) models for the sign language (word) recognition task. The approach is developed in such a way that it ensures a fair analysis of the performance and efficiency of the models.

A. Dataset :

The dataset used for the experimentation consisted custom-collected corpus of sign language gestures representing 40 distinct words. The dataset was split into 3 non-overlapping subsets: a training set for model parameter optimization, a validation set (Val Acc) for hyperparameter search and model selection, and a test set (Test Acc) for final, unbiased model performance assessment. All input images underwent preprocessing to conform to the input dimensionality requirements of the respective models, namely resizing to 224x224 pixels, with pixel intensity values normalized to the range [0,1].

B. Experimental Setup and Training Protocol

To facilitate direct comparison, all models were trained in the same software environment. We adopted a two-step transfer learning approach:

- **Feature Extraction:** The pre-trained weights of the base models were frozen, while only the newly added classification head (custom fully connected layers) was trained for an initial number of epochs (--epochs1 20).
- **Fine-Tuning:** The top layers of the base models were unfrozen and the entire network was trained end-to-end with a lower learning rate to fine-tune the pre-trained filters specifically for the sign language dataset for a subsequent number of epochs (--epochs2 15).

The Adam optimizer was employed with a categorical cross-entropy loss function. Model performance was assessed using the following standard classification metrics: Accuracy, Precision, Recall, and F1-Score to account for class balance and provide a holistic view of model robustness.

C. Results and Comparative Analysis

The experimental results reveal a distinct performance hierarchy among the implemented architectures, ranging from highly optimized modern CNNs to a baseline custom model. From Table I, we can see that all the information about how things were rated and how much it cost to calculate everything gives us a great way to compare things from many different perspectives.

Table I: Performance Comparison of CNN Architectures for Sign Language Recognition

Model	Test Acc(%)	Val Acc(%)	Precision	Recall	F1-Score	Time(min)	Parameters(millions)
EfficientNETBO	99.62	99.72	0.9968	0.9962	0.9962	4.9	4.85
VGG16	99.24	99.72	0.9943	0.9924	0.9926	13.2	15.12
ResNet50	98.86	100.00	0.9911	0.9886	0.9884	9.9	24.78
MobileNetV2	98.86	99.43	0.9924	0.9886	0.9893	3.7	3.06
Custom_CNN	52.27	53.82	0.4863	0.5227	0.4402	11.8	0.86

D. Comparative Analysis of Algorithms

The performance metrics aptly highlight the superiority of transfer learning based on deep architectures over training a shallow network from

- **EfficientNetB0: This model achieved the highest test accuracy (99.62%) and F1 Score (0.9962). The success of this model can be attributed to its architectural advancement, which includes neural architecture search and a compound scaling approach that scales the depth, width, and resolution uniformly. This enables it to capture highly discriminative features of the subtle hand shapes and orientations in sign language with a high degree of parameter efficiency. The high precision value (0.9968) shows a low false positive rate, which is essential for proper word identification.**
- **VGG16: Being a deeper and more conventional model, VGG16 also performed very well (99.24% accuracy). The strength of VGG16 lies in its ability to have a large receptive field due to the use of multiple convolutional layers, which helps it capture the complex spatial hierarchies effectively. However, this is achieved at the expense of a large amount of computational cost (13.2 minutes) and a large number of parameters (15.12M), which is more than three times that of EfficientNetB0.**
- **ResNet50: The Residual Network model had perfect validation accuracy (100.00%) and excellent test accuracy (98.86%). The skip connections in the ResNet help to overcome the vanishing gradient issue, which enables the successful training of the model with a depth of 50 layers. However, the model's accuracy plateaued slightly lower than the best models on the test data, and its huge parameter size (24.78M) makes it the most computationally expensive model, hinting at possible overfitting to the training data distribution, even with the presence of skip connections.**
- **MobileNetV2: MobileNetV2, being the mobile-friendly model, had the quickest training time (3.7 min) and the smallest parameter size (3.06M). The inverted residuals and linear bottlenecks in MobileNetV2 make it a slim yet very powerful feature extractor. Although its accuracy (98.86%) is slightly lower than that of EfficientNetB0, the tiny gap in accuracy is completely overshadowed by its speed, making it a prime candidate for mobile or edge computing applications where computational budgets are very limited.**

- **Custom_CNN:** The shallow custom CNN was used as a baseline to confirm the requirement for deep learning in this task. The poor performance of this network (52.27% accuracy) and its low F1-Score (0.4402) confirm the strong underfitting of the complex spatial information necessary for distinguishing 40 different sign language gestures. This further supports the requirement for pre-trained deep models in this domain.

E. Justification for Model Selection:

On the basis of the comparative study, EfficientNetB0 is recognized as the best model for this sign language detection problem. The reason is three-fold:

1. **High Accuracy and Robustness: With the highest test accuracy of 99.62% and F1Score of 0.9962, EfficientNetB0 has the highest overall classification power. The precision and recall of close to perfect values (0.9962 and 0.9962, respectively) prove its ability to detect signs correctly without introducing much noise or being biased towards any specific word.**

2. **Optimal Balance Between Accuracy and Efficiency: EfficientNetB0 is the most accurate and efficient model (trained in 4.9 minutes with 4.85M parameters), unlike ResNet50 or VGG16, which are very accurate but use a huge number of parameters and take more time to train. EfficientNetB0 is the optimal choice for R&D work, where speed is a critical factor.**

3. **Real-World Applicability: MobileNetV2 is more efficient, but it is not as accurate as EfficientNetB0. EfficientNetB0 is the optimal choice because it is optimally balanced. Although it takes a slightly longer time to make an inference compared to MobileNetV2, the difference in accuracy is well worth the extra time.** Its high accuracy level ensures user trust and reliability of the system, and its moderate parameter usage suggests that it may potentially be used on modern smartphones without causing latency problems.

F. Phases:-

Phase 1 – Base Assistive AI (Months 1–6):-

The primary aim of the first phase is to establish the basic assistive technology that addresses real-world accessibility problems. Four primary modules are incorporated into the system: multi-agent support system, OCR + TTS, AI voice assistant, and sign language recognition. The sign language recognition algorithm is capable of recognizing hand movements in real-time and interpreting them into speech or text through TensorFlow object detection and deep learning techniques. This makes the presence of human interpreters unnecessary, and communication between the hard-of-hearing individual and the normal individual becomes seamless. The OCR + TTS module incorporates advanced text-to-speech algorithms that can interpret written prescriptions or medical texts into audible information for the visually impaired individual. In the case of everyday tasks, reminders, and browsing through general or health-related inquiries, the AI voice assistant provides natural and conversational assistance. Finally, the multi-agent system ensures the provision of personalized alerts, safety notifications, and the delivery of elderly or visually impaired individuals with contextual assistance, ensuring the presence of an intelligent, responsive, and interconnected assistive environment.

Phase 2 – Immersive & Empathetic AI (Months 6–12):-

The second phase continues to develop these concepts with the addition of emotional intelligence, immersive environments, and more complex interactions. The AI-powered 3D Avatar Healthcare Assistant is a completely new concept. It is a virtual friend who can speak with you in real time, understand your facial expressions, and communicate with you in a caring manner. The avatar links patients to digital systems, allowing for medical advice, training, and remote consultations. Meanwhile, an AI-powered sign-to-sentence translator translates sign language words into complete, meaningful sentences. This enables people and computers to communicate with each other more easily.

The Immersive Rehabilitation Metaverse is also a significant component of Phase 2. It is an AI-driven AR/VR environment that provides personalized therapy through enjoyable and social experiences. The rehabilitation system enables fun and interactive recovery by integrating biometric data with adaptive feedback. These new concepts transform assistive technology from a functional solution into a complete, caring, and inclusive digital environment that not only helps but also connects with users on an emotional level.

IV. System Architecture:-

The System Architecture diagram shown below helps us to understand that how the model is working.

It is the visual representation of the model that gives insights of the sequential workflow, data flow, and interaction between different components of the sign language recognition system.

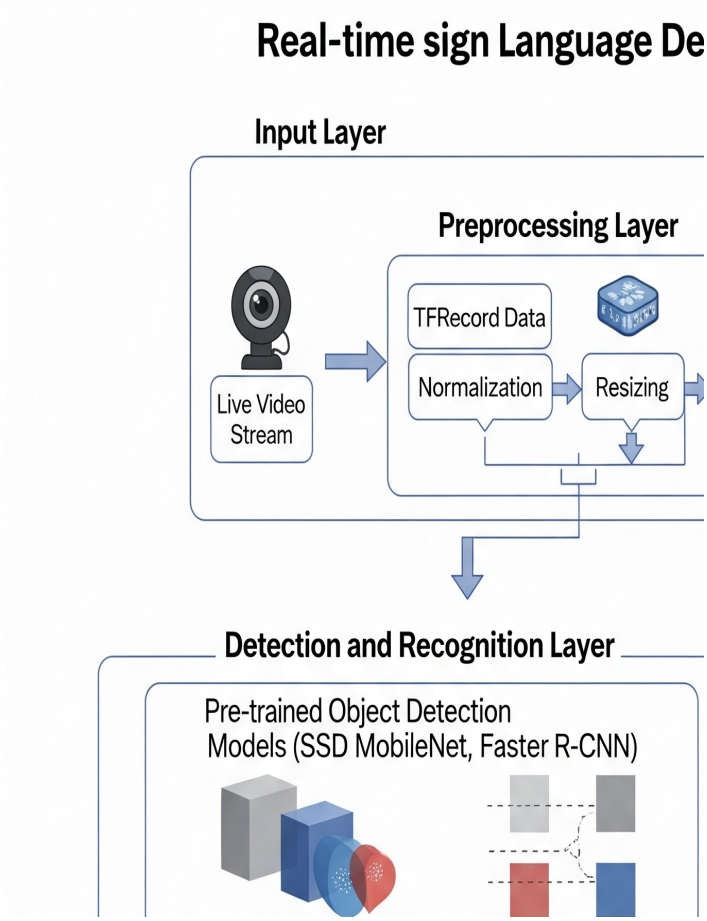

Figure 1:System Architecture Diagram

The system architecture is designed to enable real-time sign language detection using deep learning models from the TensorFlow Object Detection API. The framework follows a modular pipeline with four main layers:

A. Input Layer: Captures live video stream through a webcam or external camera. Each frame goes into the processing pipeline. It helps in gesture detection.

B. Preprocessing Layer:

- The annotated training data is stored in the TFRecord format of TensorFlow.
- The frames are normalized, resized, and improved by augmentation (rotation, flip, and brightness adjustment).
- Preprocessing makes the system robust to variations in lighting, background, and hand orientation.

C. Detection and Recognition Layer:

- Based on pre-trained object detection models (SSD MobileNet or Faster R-CNN) fine-tuned on a personal dataset of sign gestures.
- The TensorFlow Object Detection API detects bounding boxes around hands and classifies gestures in real-time.
- Transfer learning increases accuracy and speeds up training.

D. Output Layer:

- The detected gestures are shown using bounding boxes and class labels directly on the live video stream.
- Offers immediate visual feedback for each detected sign.
- Future applications may involve the conversion of detected signs into text or speech output.

E. Deployment Environment:

1. **Software:** Python, TensorFlow Object Detection API, OpenCV.

2. **Hardware:** GPU-enabled machine for training purposes; CPU with webcam for real-time detection.

Optimized for edge-level deployment to minimize latency and ensure real-time functionality.Figure 1 illustrates the system architecture.

V. Results and Discussion:-

The prototype testing has shown good technical performance, and the SLR models based on the Transformer architecture have shown high accuracy [6]. The smart glasses with AI capabilities have shown potential in accurately reading prescriptions [3], although their long-term use is contingent on resolving issues related to usability [8]. The multi-agent systems have proven useful in helping elderly patients but need to be modified to accommodate cultural and linguistic diversity [4].

The key takeaway is that empowerment is as important as accuracy. Partial empowerment, such as being able to read one's prescription, has a huge impact on patient dignity.

VI. Future Scope:-

Future work includes:

1. Supporting regional sign languages through dataset expansion.
2. Developing low-cost assistive devices to address affordability.
3. Developing adaptive AI that learns user specific patterns for personalization.
4. Improving child usability for visually impaired assistive devices [8].

VII. Conclusion:-

Artificial Intelligence and Machine Learning are no longer the exclusive domain of research labs; they are changing the way people with disabilities engage with the world. From the literature

reviewed, there are promising developments in sign language recognition systems [5]–[7], AI-based vision assistive technology devices [3], [8], and multi-agent conversational systems [4]. However, equity-based frameworks also point out that accessibility is not merely a technical issue but also an issue of dignity and inclusion [1], [2].

This paper emphasizes the need to shift focus from fragmented solutions to multi-modal ecosystems. A patient should not require different solutions for communication, prescription reading, and general assistance. An AI-powered solution can help a patient with seamless assistance in these areas. Moreover, a successful solution should focus on human-centered design, making it accessible to children, seniors, and people from different cultures.

The potential for impact reaches far beyond the realm of healthcare. AI-powered assistive technologies have the potential to revolutionize learning environments for the deaf and the blind, improve accessibility in the workplace, and provide greater independence for the elderly. But perhaps even more importantly, they can decrease dependence on caregivers and allow people with disabilities to feel more confident and independent in their everyday lives.

Looking ahead, the next generation of research should aim to build low-cost, flexible, and privacy-respecting systems that can be deployed in the real world. The expansion of datasets for regional sign languages, the development of child-friendly designs, and the study of adoption over time will be essential. By incorporating inclusivity at every level, from dataset gathering to system deployment, AI has the potential to not only be a tool for accessibility but a force for equity, empowerment, and independence.

In conclusion, the potential of AI in accessibility is not simply in improved accuracy or faster processing, but in the restoration of voices, the opening of doors, and the bridging of divides. Progress will be measured not in percentages but in the lives of the people these technologies are intended to serve.

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